

## 1. Right Angle Triangle Pattern

```
In [4]: for i in range(1,6):
        print(' * ' * i)
```

```
*
* *
* * *
* * * *
* * * * *
```

## 2. Inverted Right Angle Triangle Pattern

```
In [6]: for i in range(5,0,-1):
        print(' * ' * i)
```

```
* * * * *
* * * *
* * *
* *
*
```

## 3. Pyramid Pattern

```
In [7]: for i in range(1,6):
        print('*(5-i)+' * '*(2*i-1))
```

```
*
* * *
* * * * *
* * * * * * *
* * * * * * * *
```

## 4. Inverted Pyramid Pattern

```
In [8]: for i in range(5,0,-1):
        print('*(5-i)+' * ' *(2*i-1))
```

```
* * * * *
* * * * *
* * * *
* * *
*
```

### 5. Diamond Pattern

```
In [9]: for i in range(1,6):
        print('*(5-i)+' * ' *(2*i-1))
        for i in range(4,0,-1):
            print('*(5-i)+' * ' *(2*i-1))
```

```
*
* * *
* * * * *
* * * * * * *
* * * * * * * *
* * * * * *
* * * *
* * *
```

### 6. Hallow Square Pattern

```
In [12]: for i in range(5):
        for j in range(5):
            if i==0 or i==4 or j==0 or j==4:
                print('*',end='')
            else:
                print(' ',end='')
        print()
```

```
*****
*   *
*   *
*   *
*   *
*****
```

### 7. Full Square Pattern

```
In [13]: for i in range(5):
          print(' * '*5)
```

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

### 8. Right Angle triange(Number Pattern)

```
In [14]: for i in range(1, 6):
          print(' '.join(str(x) for x in range(1, i + 1)))
```

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

### 9. Inverted Right Angle Triangle(Number Pattern)

```
In [15]: for i in range(5,0,-1):
          print(' '.join(str(x) for x in range(1,i+1)))
```

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

## 10. Floyd's triangle

```
In [16]: num=1
         for i in range(1,6):
             for j in range(1,i+1):
                 print(num,end=' ')
                 num+=1
             print()
```

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

## 11. Hollow Right Angle Triangle

```
In [19]: for i in range(1,6):
         for j in range(1,i+1):
             if j==1 or j==i or i==5:
                 print('*',end=' ')
             else:
                 print(' ',end=' ')
         print()
```

```
*
* *
*  *
*   *
* * * * *
```

## 12. Hollow Pyramid Pattern

```
In [21]: for i in range(1, 6):
         for j in range(5 - i):
             print(' ', end=' ')
         for j in range(2 * i - 1):
             if j == 0 or j == 2 * i - 2 or i == 5:
                 print('*', end=' ')
             else:
                 print(' ', end=' ')
         print()
```

```

        else:
            print(' ', end=' ')
    print()

```

```

      *
     * *
    *   *
   *     *
  *       *
 *         *
* * * * *

```

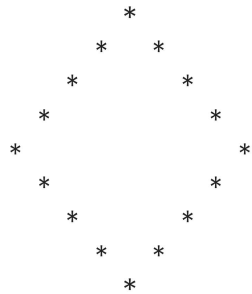
### 13. Hollow Diamond Pattern

```

In [22]: n = 5
for i in range(1, n + 1):
    for j in range(n - i):
        print(' ', end=' ')
    for j in range(2 * i - 1):
        if j == 0 or j == 2 * i - 2:
            print('*', end=' ')
        else:
            print(' ', end=' ')
    print()

for i in range(n - 1, 0, -1):
    for j in range(n - i):
        print(' ', end=' ')
    for j in range(2 * i - 1):
        if j == 0 or j == 2 * i - 2:
            print('*', end=' ')
        else:
            print(' ', end=' ')
    print()

```



#### 14. Hallow Diamond (Number Pattern)

```
In [23]: n = 5
for i in range(1, n + 1):
    for j in range(n - i):
        print(' ', end=' ')
    for j in range(2 * i - 1):
        if j == 0 or j == 2 * i - 2:
            print(i, end=' ')
        else:
            print(' ', end=' ')
    print()

for i in range(n - 1, 0, -1):
    for j in range(n - i):
        print(' ', end=' ')
    for j in range(2 * i - 1):
        if j == 0 or j == 2 * i - 2:
            print(i, end=' ')
        else:
            print(' ', end=' ')
    print()
```

```

      1
    2 2
  3   3
4    4
5    5
  4   4
    3 3
      2 2
        1

```

### 15. Butterfoly Pattern

```

In [25]: n = 5
for i in range(1, n + 1):
    for j in range(1, i + 1):
        print(j, end=' ')
    for j in range(2 * (n - i)):
        print(' ', end=' ')
    for j in range(1, i + 1):
        print(j, end=' ')
    print()
for i in range(n, 0, -1):
    for j in range(1, i + 1):
        print(j, end=' ')
    for j in range(2 * (n - i)):
        print(' ', end=' ')
    for j in range(1, i + 1):
        print(j, end=' ')
    print()
n = 5
for i in range(1, n + 1):
    for j in range(i):
        print('*', end=' ')
    for j in range(2 * (n - i)):
        print(' ', end=' ')
    for j in range(i):
        print('*', end=' ')
    print()
for i in range(n, 0, -1):
    for j in range(i):
        print('*', end=' ')

```

```

for j in range(2 * (n - i)):
    print(' ', end=' ')
for j in range(i):
    print('*', end=' ')
print()

```

```

1                1
1 2              1 2
1 2 3            1 2 3
1 2 3 4          1 2 3 4
1 2 3 4 5 1 2 3 4 5
1 2 3 4 5 1 2 3 4 5
1 2 3 4          1 2 3 4
1 2 3            1 2 3
1 2              1 2
1                1
*                *
* *              * *
* * *            * * *
* * * *          * * * *
* * * * * * * * * *
* * * * * * * * * *
* * * *          * * * *
* * *            * * *
* *              * *
*                *

```

## 16. Hollow Number Pyramid

```

In [26]: n = 5
for i in range(1, n + 1):
    for j in range(n - i):
        print(' ', end=' ')

    for j in range(1, 2 * i):
        if j == 1 or j == 2 * i - 1 or i == n:
            print(i, end=' ')
        else:
            print(' ', end=' ')
    print()

```



```

      1
     2 2
    3   3
   4     4
  5 5 5 5 5 5 5 5

```

### 17. Full Star Pyramid

```

In [27]: n = 5

for i in range(1, n + 1):

    for j in range(n - i):
        print(' ', end=' ')

    for j in range(2 * i - 1):
        print('*', end=' ')

    print()

```

```

      *
     * * *
    * * * * *
   * * * * * * *
  * * * * * * * *
 * * * * * * * * *

```

### 18. Inverted Full Star Pyramid

```

In [28]: n = 5

for i in range(n, 0, -1):

    for j in range(n - i):
        print(' ', end=' ')

    for j in range(2 * i - 1):
        print('*', end=' ')

    print()

```

```

* * * * *
 * * * * *
  * * * *
   * * *
    *

```

### 19. Left Aligned Pyramid Pattern

```

In [29]: n = 5

for i in range(1, n + 1):

    for j in range(i):
        print('*', end=' ')
    print()

n = 5

for i in range(1, n + 1):
    for j in range(1, i + 1):
        print(j, end=' ')
    print()

```

```

*
* *
* * *
* * * *
* * * * *
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

```

### 20. Right Aligned Pyramid Pattern

```

In [30]: n = 5

for i in range(1, n + 1):

```

```

    for j in range(n - i):
        print(' ', end=' ')

    for j in range(1, i + 1):
        print(j, end=' ')

    print()

n = 5

for i in range(1, n + 1):

    for j in range(n - i):
        print(' ', end=' ')

    for j in range(i):
        print('*', end=' ')

    print()

```

```

      1
     1 2
    1 2 3
   1 2 3 4
  1 2 3 4 5
    *
   * *
  * * *
 * * * *
* * * * *

```

In [ ]: